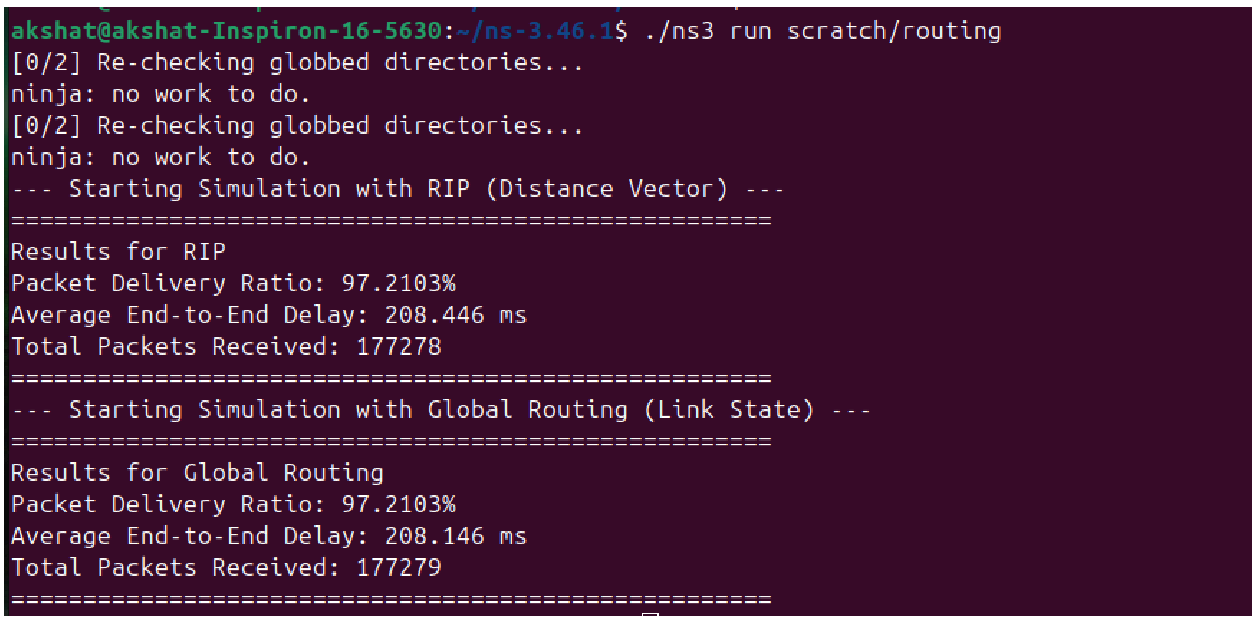
<u>Homework -7</u>

Ans 1)
Results for RIP
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.446 ms
Total Packets Received: 177278

Results for Global Routing
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.146 ms
Total Packets Received: 177279



```
akshat@akshat-Inspiron-16-5630:~/ns-3.46.1$ ./ns3 run scratch/routing
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.
--- Starting Simulation with RIP (Distance Vector) ---
=====================================================
Results for RIP
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.446 ms
Total Packets Received: 177278
=====================================================
--- Starting Simulation with Global Routing (Link State) ---
=====================================================
Results for Global Routing
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.146 ms
Total Packets Received: 177279
=====================================================
```

Ans 2)

Results for RIP

Packet Delivery Ratio: 41.7496%

Average End-to-End Delay: 211.787 ms

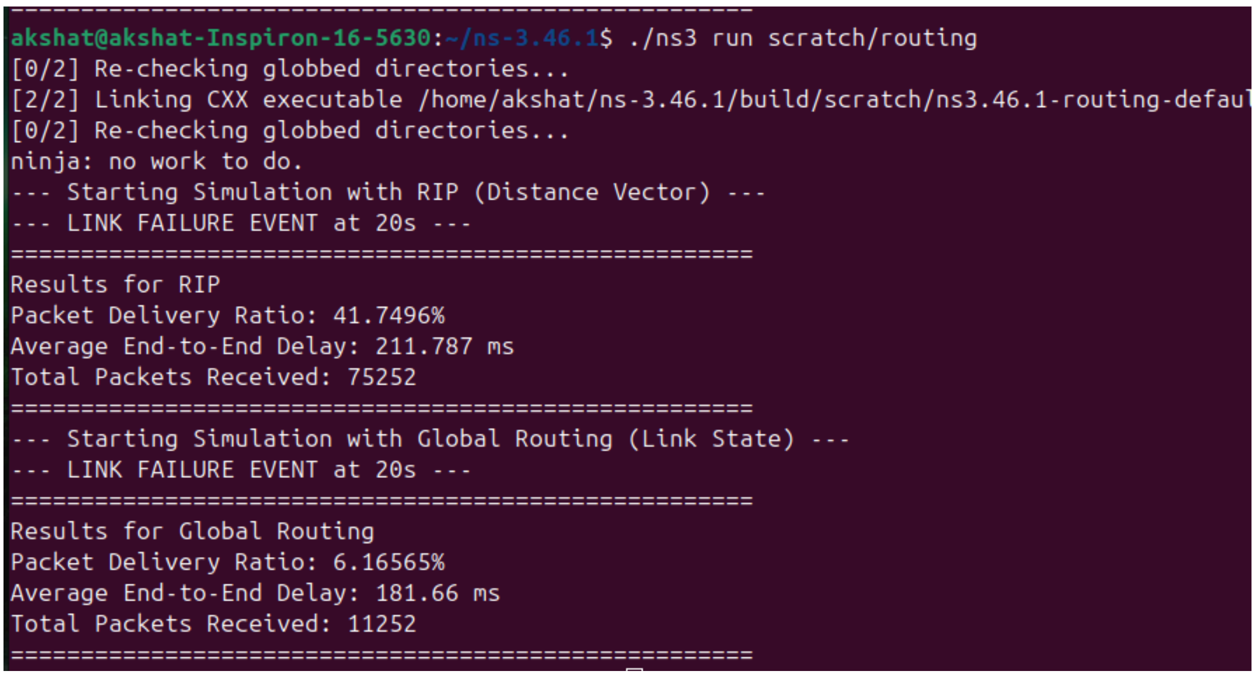
Total Packets Received: 75252

Results for Global Routing

Packet Delivery Ratio: 6.16565%

Average End-to-End Delay: 181.66 ms

Total Packets Received: 11252



```
=====================================================
akshat@akshat-Inspiron-16-5630:~/ns-3.46.1$ ./ns3 run scratch/routing
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home/akshat/ns-3.46.1/build/scratch/ns3.46.1-routing-defaul
[0/2] Re-checking globbed directories...
ninja: no work to do.
--- Starting Simulation with RIP (Distance Vector) ---
--- LINK FAILURE EVENT at 20s ---
=====================================================
Results for RIP
Packet Delivery Ratio: 41.7496%
Average End-to-End Delay: 211.787 ms
Total Packets Received: 75252
=====================================================
--- Starting Simulation with Global Routing (Link State) ---
--- LINK FAILURE EVENT at 20s ---
=====================================================
Results for Global Routing
Packet Delivery Ratio: 6.16565%
Average End-to-End Delay: 181.66 ms
Total Packets Received: 11252
=====================================================
```

Ans 3)

Time instant	**Gateway for Global Routing Protocol**	**Gateway for RIP**
10 sec	**10.0.2.2**	**10.0.2.2**
60 sec	**10.0.2.2**	**10.0.2.2**
120 sec	**10.0.2.2**	**10.0.2.2**
180 sec	**10.0.2.2**	**10.0.2.2**
240 sec	**10.0.2.2**	**10.0.7.2**
300 sec	**10.0.2.2**	**10.0.7.2**

So, from theory, we know that global routing is a static link state routing protocol, so it computes all routes once at the start, using the entire topology graph, and the routes decided never change, even if there is a link failure, whereas RIP (Routing Information Protocol) is a dynamic protocol wherein routers exchange routing information with neighbours in the graph topology after every fixed interval and thus can detect link failures, if they happen.

As we see in the table, the global routing protocol always uses the gateway "10.0.2.2".This remained unchanged even when there was a link failure between node 2 and node 3. As a result, the route becomes invalid: although the next-hop IP (10.0.2.2) still appears in the table, node 2 can no longer forward packets to that interface because of a link failure between node 2 and 3. So the packets that were forwarded along this path were all dropped. That is why we see a very small packet delivery ratio of 6.16565% in comparison to the RIP protocol, which has a packet delivery ratio of 41.7496%.

The reason is if we look at the RIP, at timepoints +10s / +60s / +120s / +180s, the gateway remains unchanged and stays 10.0.2.2, but at time instant 240 sec and 300 sec, the gateway for 10.0.3.2 changed to "10.0.7.2", this indicates that upon detecting link failure between 2 and 3, the nodes 2 and 3 no longer shared information with each others as neigbours and thus RIP switched to an alternated path for sending data from source node 1 to destination node 6. This alternate path goes through node 4.

We have attached to our submission the 2 routing files that we have obtained in this case.

Ans 4)

```
akshat@akshat-Inspiron-16-5630:~/ns-3.46.1$ ./ns3 run scratch/routing
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home/akshat/ns-3.46.1/build/scratch/ns3.46.1-routing-default
[0/2] Re-checking globbed directories...
ninja: no work to do.
--- Starting Simulation with RIP (Distance Vector) ---
--- LINK FAILURE EVENT at 20s ---
=====================================================
Results for RIP
Packet Delivery Ratio: 97.2096%
Average End-to-End Delay: 220.347 ms
Total Packets Received: 176736
=====================================================
--- Starting Simulation with Global Routing (Link State) ---
--- LINK FAILURE EVENT at 20s ---
=====================================================
Results for Global Routing
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 219.832 ms
Total Packets Received: 177273
=====================================================
```

This was the output that we obtained after adding two new nodes, 7 and 8, and also assigning the IPs between the new links as stated in the Figure. The output clearly shows that the packet delivery ratio for both protocols has increased to approximately 97.2.

The reason why there is so a high packet delivery ratio for both these protocols is that in the earlier version, before the link failure happened, the shortest path both the protocols computed to reach node 6 from node 1 included the link from node 2 to node 3, and this argument was even supported by the 2 ".route" files that we obtained in that case. And when the link between node 2 and node 3 failed, there was too much data loss along that path. But now, the addition of 2 nodes, 7 and 8, no longer keeps the shortest path in the network from node 1 to node 6 to go through the link between node 2 and node 3. That is why even after the link failure happened, neither of them ever required to change the table, as no data was being sent along that link.

This claim is even supported by the two ".route" files that we obtained in this case as well.

1) In the "rip-routing.routes", we see the following:

```
akshat@akshat-Inspiron-16-5630:~/ns-3.46.1$ cat rip-routing.routes
Node: 2, Time: +10s, Local time: +10s, IPv4 RIP table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
10.0.9.0        10.0.2.2        255.255.255.0   UGS   3      -      -   2
10.0.3.0        10.0.7.2        255.255.255.0   UGS   4      -      -   3
10.0.6.0        10.0.7.2        255.255.255.0   UGS   3      -      -   3
10.0.5.0        10.0.7.2        255.255.255.0   UGS   2      -      -   3
```

This clearly indicates that for any host in "10.0.3.0/24" or "10.0.6.0/24"(these are the destination host used by the traffic generator in the simulation), node 2 will send packets to gateway "10.0.7.2", that is through node 4 and thus even if link failure happened between node 2 and node 3 data still continued to go through node 4 to node 6 as it remained the shortest path ever.

2) In the “global-routing.routes”, we see the following:

```
  Priority: -10 Protocol: ns3::Ipv4GlobalRouting
Node: 2, Time: +10s, Local time: +10s, Ipv4GlobalRouting table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
10.0.1.1        10.0.1.1        255.255.255.255 UH    -      -      -   1
10.0.2.2        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.8.1        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.5.1        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.7.2        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.8.2        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.9.1        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.5.2        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.6.1        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.9.2        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.3.1        10.0.2.2        255.255.255.255 UH    -      -      -   2
10.0.3.2        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.6.2        10.0.7.2        255.255.255.255 UH    -      -      -   3
10.0.1.0        10.0.1.1        255.255.255.0   UG    -      -      -   1
10.0.2.0        10.0.2.2        255.255.255.0   UG    -      -      -   2
10.0.8.0        10.0.2.2        255.255.255.0   UG    -      -      -   2
10.0.9.0        10.0.2.2        255.255.255.0   UG    -      -      -   2
10.0.3.0        10.0.2.2        255.255.255.0   UG    -      -      -   2
10.0.5.0        10.0.7.2        255.255.255.0   UG    -      -      -   3
10.0.7.0        10.0.7.2        255.255.255.0   UG    -      -      -   3
10.0.6.0        10.0.7.2        255.255.255.0   UG    -      -      -   3
10.0.3.0        10.0.7.2        255.255.255.0   UG    -      -      -   3
```

This also clearly indicates what we observed in the case of RIP. For any host in ”10.0.3.2” or “10.0.6.2”, node 2 will send packets to the gateway “10.0.7.2”, that is, through node 4, and hence, as explained above, there will be no packet drops as the shortest path selected from start never contained the link which failed, i.e, between node 2 and node 3.

We have attached to our submission the 2 routing files that we have obtained in this case.

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